

महाराष्ट्र वन सेवा सुख परीक्षा - 2025
दि. 5 तै 9 से, 2026



2025

H24

BOOKLET NO.

100304

Forest Services
Agricultural Engineering

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional /Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note –1. Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.



SECTION – A

Q1.	Solve any five out of seven.	(8×5=40)
Q1.	(a) Enlist different watershed management objectives.	8
Q1.	(b) Define sensor resolution and describe different types of resolution.	8
Q1.	(c) Define subsurface drainage. Give general consideration while planning effective subsurface drainage system.	8
Q1.	(d) What are the objectives of poultry house ? Describe different types of poultry houses used commercially in India.	8
Q1.	(e) Define hydrograph and discuss components of storm hydrograph.	8
Q1.	(f) Write short note on different irrigation indices.	8
Q1.	(g) Define concrete. Enlist the reasons why concrete has attained the status of a major building material. Also, enlist the properties of concrete.	8
Q2.	(a) Define water erosion and discuss factors affecting water erosion.	15
Q2.	(b) What are the types of geospatial data ? Enlist advantages and constraints of raster and vector data.	15
Q2.	(c) Discuss different soil moisture constants.	10
Q3.	(a) Discuss different types of filters in micro irrigation system.	15
Q3.	(b) Enlist the benefits of protected cultivation. Describe the factors to be considered while constructing a green house.	15
Q3.	(c) On 4 hectare land with 24% slope, the broad-base bench terrace is desired to construct for growing the crops. During 10-years recurrence interval the maximum rainfall intensity was recorded as 15cm/h for the time equal to 'Te'.	10
	Calculate :	
	1) Peak run-off rate to be disposed off.	
	2) Horizontal spacing between terrace.	
	3) Length per hectare.	
	4) Total length of terrace.	
	Assume C as 0.35.	
Q4.	(a) Discuss different types of agronomical control measures of soil erosion.	15
Q4.	(b) Define drip irrigation. Enlist its merits and constraints. Also, explain the step-by-step procedure for the design of a drip irrigation system.	15
Q4.	(c) Discuss different bulk grain storage structures commonly used in India.	10

**SECTION – B**

- Q5.** Solve **any five** out of seven. **(8×5=40)**
- Q5.** (a) Classify the different types of farm implements based on tillage operations. Differentiate between primary and secondary tillage with suitable examples. **8**
- Q5.** (b) Define a bio-diesel. Discuss the process of transesterification for diesel production from vegetable oil. **8**
- Q5.** (c) Define drying. Write a formulae for moisture content on wet basis and dry basis. A two ton lot of wheat has 22% moisture content (wet basis), dry it to 13% moisture content (dry basis).
Calculate :
- i) Bone-dry weight
 - ii) Water evaporated
 - iii) Final weight **8**
- Q5.** (d) What is a microprocessor ? Explain its application in data acquisition and control of agricultural engineering processes. **8**
- Q5.** (e) Differentiate between the following. **(4×2=8)**
- 1) Rotavator and Cultivator
 - 2) Deep tillage and sub soiling
 - 3) Tractor and power tiller
 - 4) Two stroke and four stroke engine
- Q5.** (f) Explain the working of a photovoltaic system. State its main components and agricultural applications. **8**
- Q5.** (g) Define homogenization. Why is it necessary in milk processing ? **8**
-
- Q6.** (a) Explain the constructions and working of the cooling and lubrication systems of an Internal Combustion (IC) Engine. **15**
- Q6.** (b) Explain the process of gasification of biomass. How is it utilized for running Internal Combustion (IC) engine and electric power generation ? **15**
- Q6.** (c) State the formulae to express the capacity and power requirement of bucket elevators. Explain its basic constructions. **10**



- Q7. (a)** Explain Post Harvest Technology (PHT) in detail with scope, objectives and significance with its fundamental unit operations in processing. Enlist the major causes of PHT losses. **15**
- Q7. (b)** Define multimedia and Audio-visual aids. Classify Audio-Visual aids with suitable examples. Explain the significance and advantages of using multimedia and audio-visual aids in agricultural extension and education. **15**
- Q7. (c)** Enlist types of earth moving machinery used in land development. Explain its construction and working principle of a bulldozer. Compare fixed and operating cost. **10**
-
- Q8. (a) 1)** Explain the fundamental physical principle behind centrifugal cream separation in dairy processing. State the equation for the velocity separation and explain how the size of the fat globule milk viscosity and density difference influence the efficiency of separation process. **7**
- 2)** A continuous centrifugal cream separator processes 2500 kg/hr of whole milk containing 4.2% fat. The separator is adjusted to produce cream with content of 35%. If the resulting skim milk contains 0.1% fat, calculate the exact hourly yield (in kg) of both cream and the skim milk. **8**
- Q8. (b)** Define theoretical field capacity, effective field capacity and field efficiency. A tractor is operating a 5 bottom disc plough with each bottom having width of 25cm at forward speed of 6 kmph. The Field efficiency of operation is 80%. The specific soil resistance is 0.6 kg/cm^2 and the depth of ploughing is 15cm. **15**
- Calculate :
- 1) Effective Field capacity in ha/hr.
 - 2) The total draft of the plough in kg.
 - 3) Drawbar Horsepower (DBHP)
- Q8. (c)** Define Bio-Energy. Differentiate between Biodiesel and Bio-hydrogen in terms of their production sources and characteristics. Briefly explain their scope and potential applications as alternative fuels for running Internal Combustion (IC) Engine in the agriculture sector. **10**
-